

Quantification of Konghanzheng by FIFA Rankings under Generalized Linear Mixed-effects Regression Model

Noah Kan • Statistics Department• Cal State Monterey Bay
nkan@csumb.edu



Background

Results

- Soccer is the most popular sport across continents, and there are international rivalries in many regions of the world.
- In Eastern Asia, there are intense soccer rivalries among China, Japan, and South Korea.
- “Kong-han-zheng” (恐韩症) is a popular word in China which literally means the “symptom of fearing Korea” in Chinese.
- It has become a popular term due to the strongly skewed results against China since 1978 (2 wins, 13 draws, and 23 losses as of June 2024).
- Physical ability and skills are direct factors of sports performance, and researchers have also studied that psychological factors (such as stress, anxiety, emotional pressure) and other factors (such as opponent's level, reputation, and status) affect player's biological response and performance in both individual and team sports.1-5
- National soccer teams are ranked monthly by the FIFA based on observed game outcomes, and the rank must be the strongest predictor of the probability of winning.

Objective

Whether any skewed results between rivalries are purely due to ranking or not has not been explained using systematic analysis. To this end, the objective of this statistical project is to investigate if skewed results (e.g., Kong-han-zheng) can be detected beyond observed explanatory variables (e.g., ranking of the team and of the opponent and home vs. away) by statistical modeling.

Method

- Historical data of international soccer matches and FIFA rankings were freely available at Kaggle.7-8
- All games played by China (326 games), Japan (421 games), and South Korea (326 games) were curated, and FIFA rankings of the three countries and their opponents were matched to the nearest date prior to each game.
- For each of the three countries, we used simple logistic regression to visualize the probability of winning with respect to the opponent's rank.
- For each of the three countries, we then used the following generalized linear mixed-effects regression (GLMER) for the binomial data (repeated matches per opponent).
 - Let θ_{ij} denote the probability of winning for game j against opponent i.
 - $\theta_{ij} = \exp(\mu_{ij}) / (1 + \exp(\mu_{ij}))$
 - $\mu_{ij} = \beta_0 + \beta_1 X_{1ij} + \beta_2 X_{2ij} + \beta_3 X_{3ij} + \epsilon_i$
 - X_{1ij} = FIFA ranking (China, Japan, or South Korea) prior to game j against opponent i (in 10's)
 - X_{2ij} = FIFA ranking of opponent i prior to game j (in 10's)
 - X_{3ij} = home game (1) or away game (0)
 - $\epsilon_i \sim N(0, \sigma)$ is the random-effect (repeated matches against opponent i)
- We used the same approach to analyze all countries' matches world wide (265 countries; 44353 games).

Figure 1. The fitted logistic regression model for the probability of winning (China)

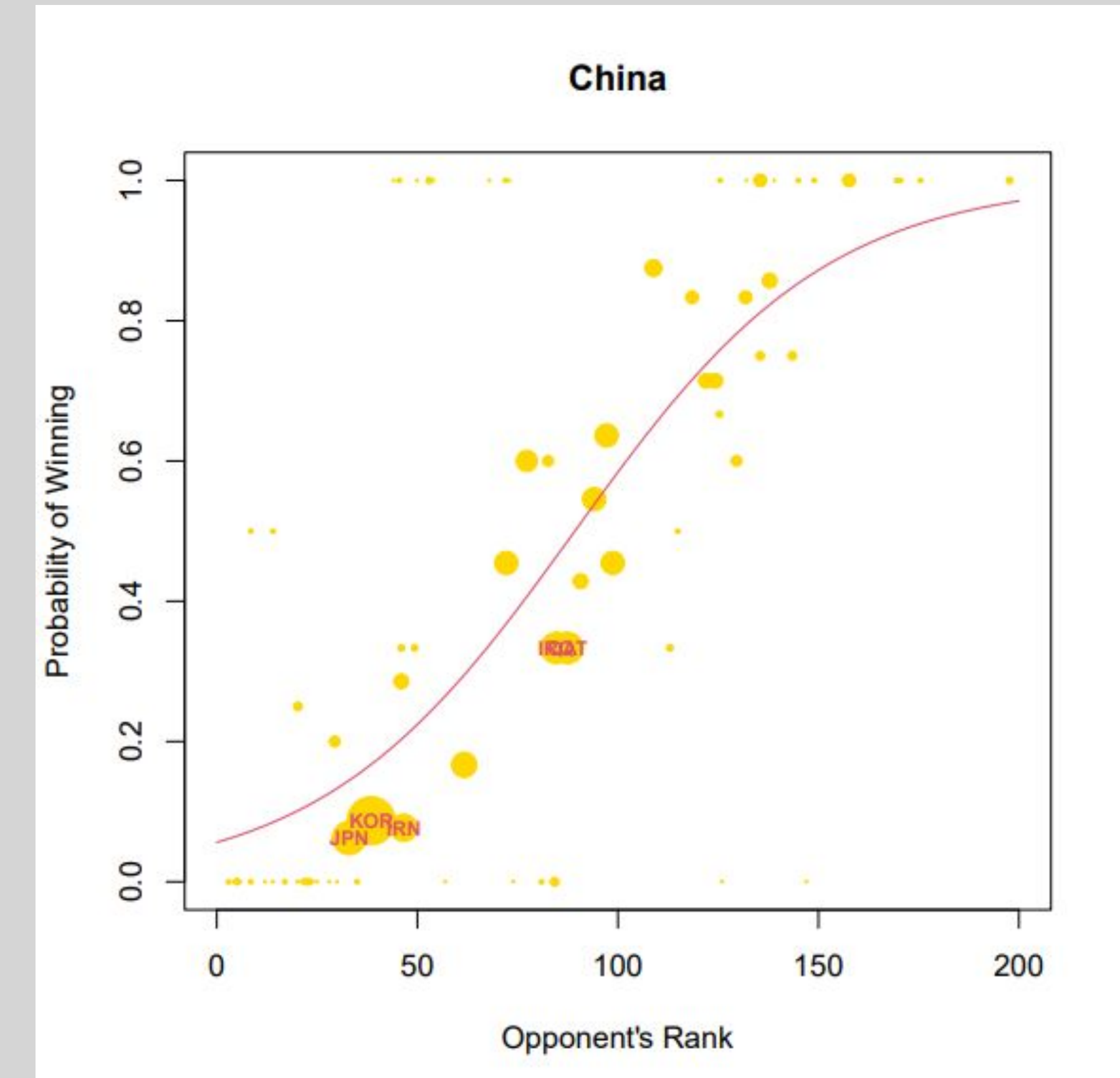


Figure 2. The fitted logistic regression model for the probability of winning (Japan)

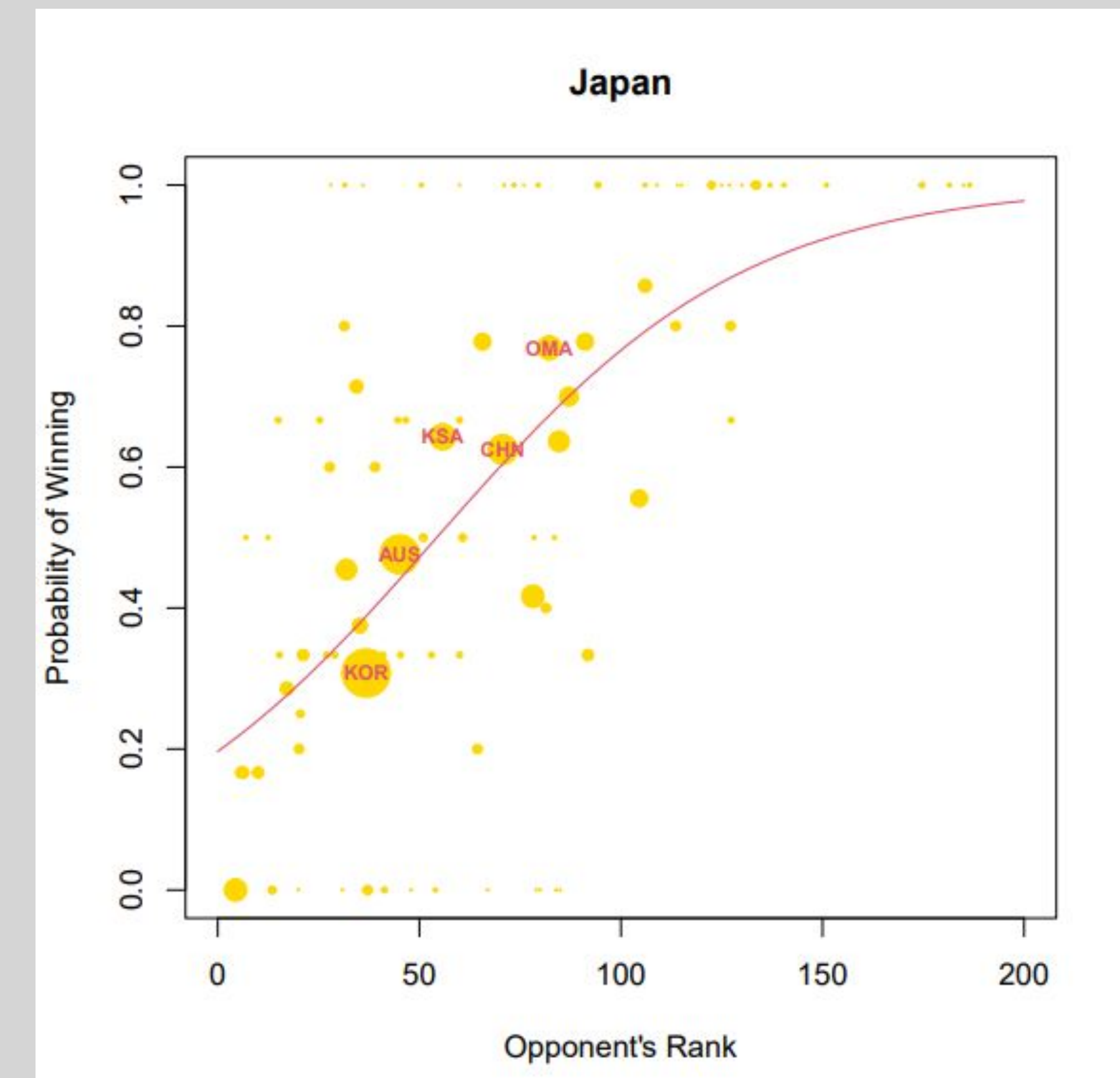


Figure 3. The fitted logistic regression model for the probability of winning (Korea)

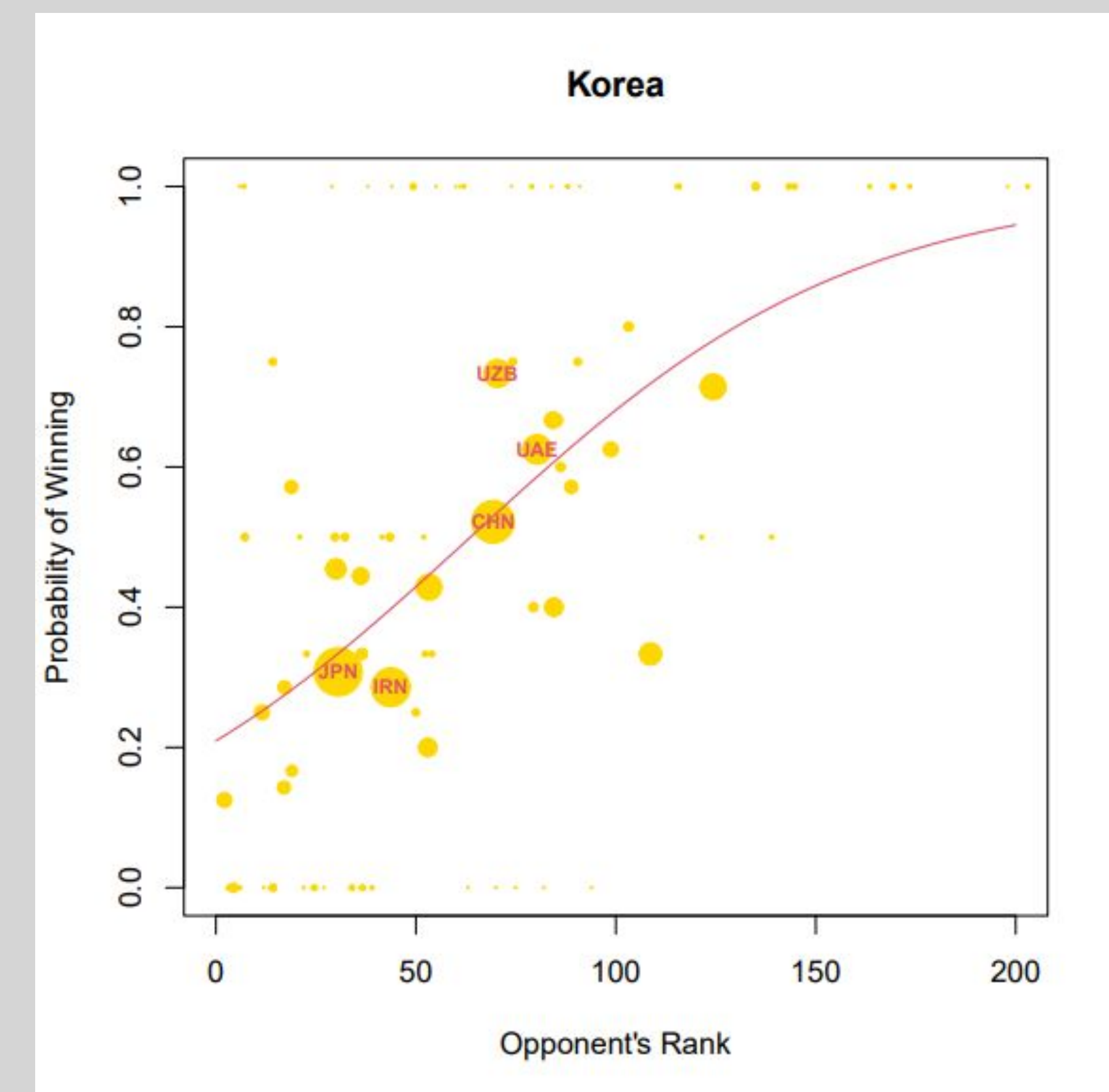


Figure 4. The random effects of opponents estimated by the GLMER for China.

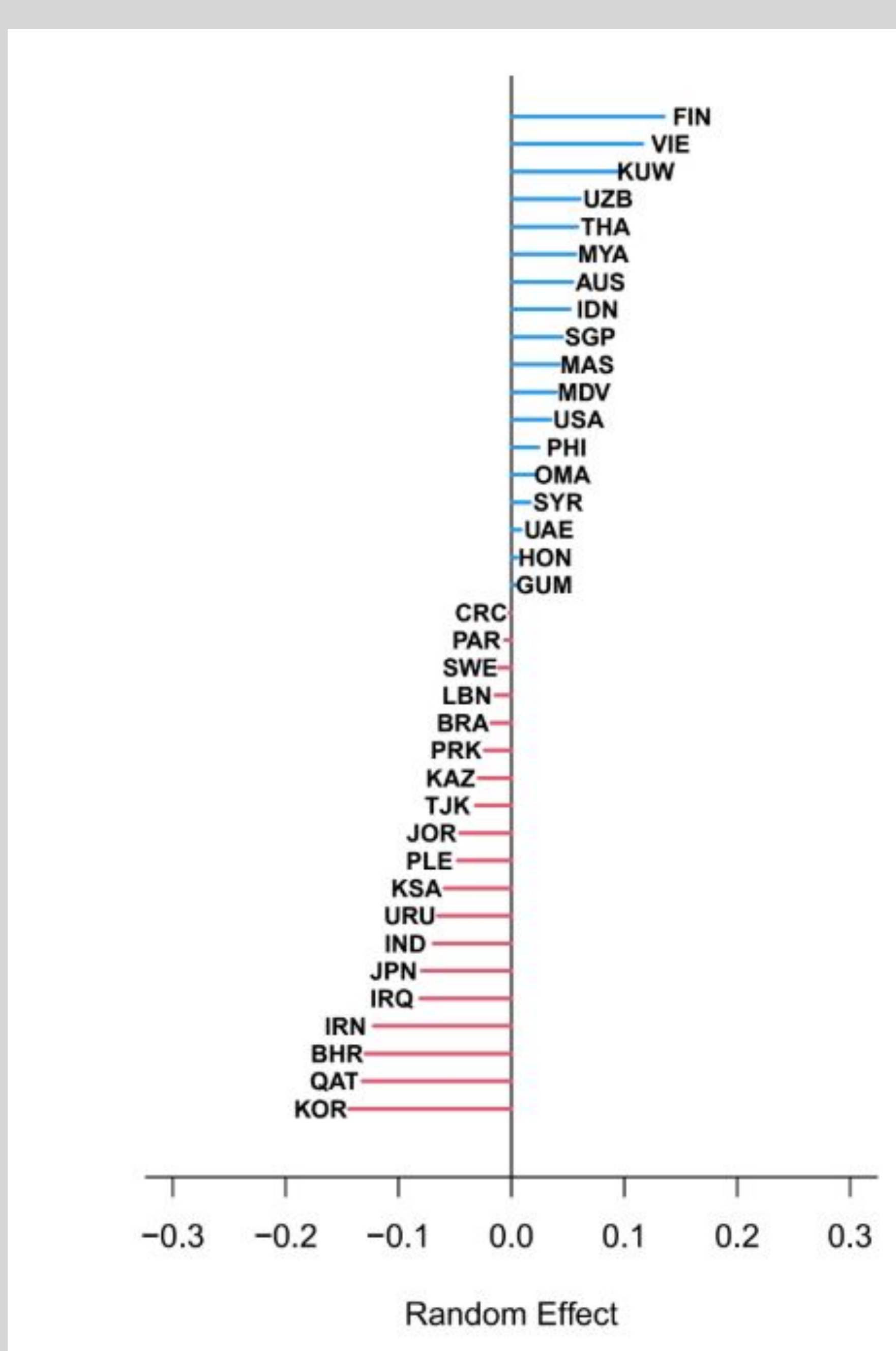


Table 1. The fixed effects of observed variables estimated by the GLMER for each country (β_0 countries rank, β_1 opponent rank + β_2 home or away indicator)

		Estimate	SE	Z-value	OR (95% CI)	P-value
China	β_0	-2.033	0.749	-2.714	NA NA	NA
	β_1	-0.101	0.100	-1.013	0.904 (0.71, 1.10)	0.311
	β_2	0.257	0.036	7.06	1.293 (1.22, 1.36)	<0.001
	β_3	1.071	0.271	3.958	2.918 (2.39, 3.45)	<0.001
Japan	β_0	-1.836	0.362	-5.065	NA NA	NA
	β_1	0.071	0.082	0.87	1.074 (0.91, 1.23)	0.385
	β_2	0.235	0.032	7.455	1.265 (1.20, 1.33)	<0.001
	β_3	0.668	0.218	3.062	1.951 (1.52, 2.38)	0.002
Korea	β_0	-1.816	0.435	-4.177	NA NA	NA
	β_1	0.039	0.090	0.439	1.04 (0.86, 1.22)	0.661
	β_2	0.197	0.029	6.799	1.218 (1.16, 1.28)	<0.001
	β_3	0.864	0.218	3.954	2.372 (1.94, 2.80)	<0.001

- For all three countries, the probability of winning clearly increases with respect to the rank of the opponent (Figures 1-3), and the odds of winning increases by 1.2 to 1.3 per 10 ranks(?) of the opponent (Table 1).
- For all three countries, the odds of winning increases by 2 to 3 when the game is at home given the ranks are the same (Table 1).
- The dispersion of random-effect (σ) could be estimated for China (Figure 2), but not for Japan and South Korea. In other words, the probability of winning could be all explained by the observable variables under the model for Japan and South Korea, but unobservable factors might impact China's outcome.
- China has not performed particularly well against Korea according to the estimated random-effect (Figure #), and it is indication that Kong-han-zheng may truly exist.

Conclusions

- Konghanzheng is not just a rumor it is proved to be true and that China tends to lose to Korea
- When looking at the world-wide results we can see a high disparity between ranks of certain countries as some countries have inflated ranks of they play less games
- Looking at the world-wide teams matchups show the same teams such as the USA have very high random effects and this can be associated to being over ranked or unde ranked.
- Random Effect can be used to analyze other countries and leagues. We can expand this method to allow for coaches to understand their teams and leagues better.
- Random-effects could be used to understand the data beyond unobservable variables.
- Mixed effect models can be used to analyze test scores, amount of crops that grow, animal populations, and many other contexts

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